

JAVASCRIPT #1

JavaScript

- Server-side scripts:

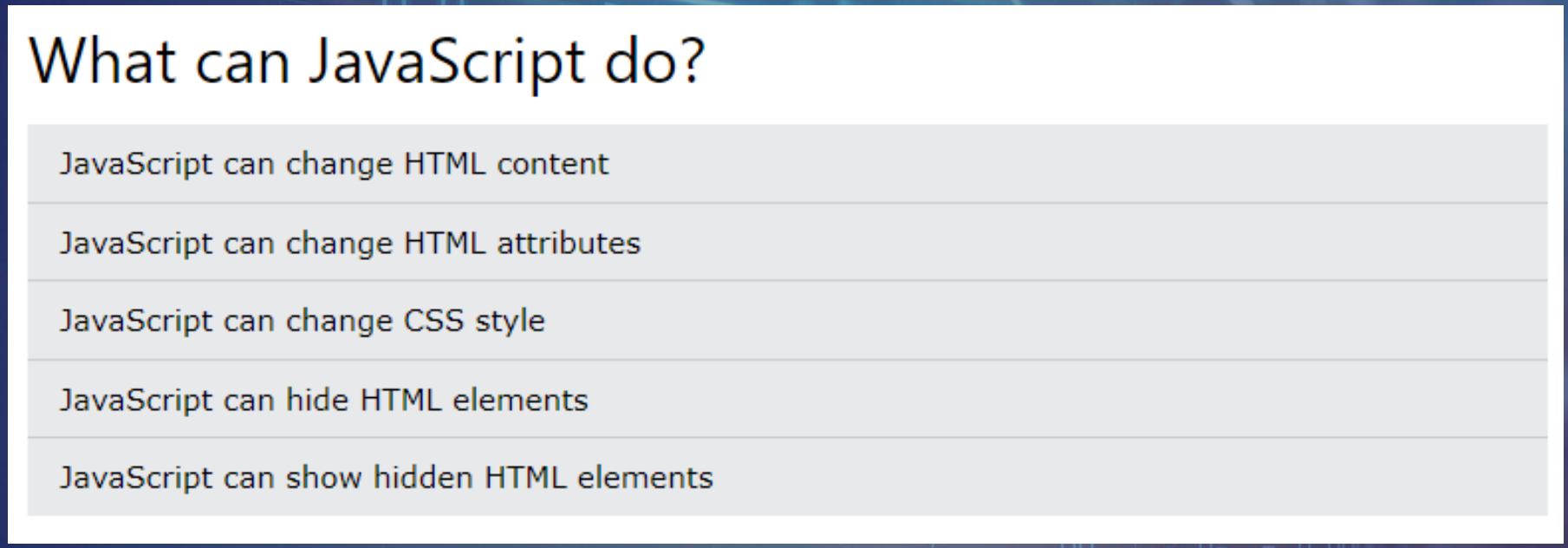
  Scripts that are run on a Web server

- Client-side script:

  Scripts that are run in the browser

WHAT CAN JAVASCRIPT DO?

some examples of what JavaScript can do.



https://www.w3schools.com/js/js_examples.asp

WHERE TO INSERT JAVASCRIPT

Where to Insert JavaScript

- JavaScript in <head>
- JavaScript in <body>
- JavaScript in an external file
- JavaScript in an external url
- JavaScript in an external folder

https://www.w3schools.com/js/js_examples.asp

WHERE TO INSERT JAVASCRIPT

In HTML, JavaScript code is inserted between <script> and </script> tags.

```
<script>
    document.getElementById("demo").innerHTML = "My JavaScript";
</script>
```

https://www.w3schools.com/js/js_whereto.asp

WHERE TO INSERT JAVASCRIPT

JavaScript in <head> or <body>

```
<head>
<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Paragraph changed.";
}
</script>
</head>
```

```
<body>
<button type="button" onclick="myFunction()">Try it</button>
<script>
function myFunction() {
  document.getElementById("demo").innerHTML = "Paragraph changed.";
}
</script>
</body>
```

https://www.w3schools.com/js/js_whereto.asp

WHERE TO INSERT JAVASCRIPT

JavaScript in external file

```
<script src="myScript1.js"></script>
<script src="myScript2.js"></script>
```

JavaScript in external URL

```
<script src="https://www.w3schools.com/js/myScript.js"></script>
```

JavaScript in external folder

```
<script src="/js/myScript.js"></script>
```

https://www.w3schools.com/js/js_whereto.asp

JAVASCRIPT OUTPUT

JavaScript can "display" data in different ways:

- Writing into an HTML element, using .innerHTML.
- Writing into the HTML output using document.write().
- Writing into an alert box, using alert().
- Writing into the browser console, using console.log().

https://www.w3schools.com/js/js_output.asp

JAVASCRIPT STATEMENTS

Values, Operators, Expressions, Keywords, and Comments.

- Semicolons separate JavaScript statements.
- statements can be grouped together in code

  blocks, inside curly brackets {...}.
- Reserved Words

https://www.w3schools.com/js/js_statements.asp

JAVASCRIPT SYNTAX

```
// How to create variables:
const PI = 3.141592653589793;
var x;
let y;

// How to use variables:
x = 5;
y = 6;
let z = x + y;
Sum = (5+6) *10

let  Fullname = 'John' + " " + "Doe"
```

https://www.w3schools.com/js/js_syntax.asp

JAVASCRIPT DATATYPE

```
// Numbers:
let length = 16;
let weight = 7.5;
// Strings:
let color = "Yellow";
let lastName = "Johnson";
// Booleans
let x = true;
let y = false;
// Object:
const person = {firstName:"John", lastName:"Doe"};
const person = {
                    firstName:"John",
                    lastName:"Doe“
                };
// Array object:
const cars = ["Saab", "Volvo", "BMW"];

// Date object:
const date = new Date("2022-03-25");
```

JAVASCRIPT CONDITIONAL

```
const time = new Date().getHours();
let greeting;

if (time < 10) {
   greeting = "Good morning";
} else if (time < 20) {
   greeting = "Good day";
} else {
   greeting = "Good evening";
}
console.log(greeting)
```

https://www.w3schools.com/js/js_if_else.asp

JAVASCRIPT FUNCTION

```
function greet(name) {
  return `Hi ${name}`;
}

const greet = (name) => {
  return `Hi ${name}`;
};

let message = greet();
```

```
// Before Arrow Function
const multiply1 = function (a,
b) {
  return a * b
}

// Arrow Function
const multiply2 = (a, b) => a
* b;

console.log(multiply1(2, 3)); //
Output: 6
console.log(multiply2(2, 3)); //
Output: 6
```

https://www.w3schools.com/js/js_function_intro.asp

HTML DOM API

The DOM API provides us with the ability to:

- Find and select elements
- Change element content and attributes
- Add, remove, or modify elements
- Change CSS styles
- Add event listeners to react to user input

Method	Description
document.getElementById(id)	Find an element by element id
document.getElementsByTagName(name)	Find elements by tag name *document.getElementsByTagName("p");*
document.getElementsByClassName(name)	Find elements by class name
document.querySelector(selector)	Find the first element that matches a CSS selector
document.querySelectorAll(selector)	Find all elements that match a CSS selector

https://www.w3schools.com/js/js_htmldom_methods.asp

JAVASCRIPT EVENTS

Common HTML Events

```
<button onclick="displayDate()">The time is?</button>
```

onchange	An HTML element has been changed
onclick	The user clicks an HTML element
onmouseover	The user moves the mouse over an HTML element
onmouseout	The user moves the mouse away from an HTML element
onkeydown	The user pushes a keyboard key
onload	The browser has finished loading the page

https://www.w3schools.com/js/js_events.asp

JAVASCRIPT EVENT LISTENER

ลดการใช้หน่วยความจำด้วยการผูก Event Listener ที่ Element

```
<div class="container">
    <div id="box">
        Hover over me!
    </div>
</div>
```

```
<script>
    const box = document.getElementById("box");
    // Let box listen for click
    box.addEventListener("click", function () {
        alert("Box clicked!");
    });
    // Let box listen for mouseover
    box.addEventListener("mouseover", function () {
        box.innerHTML = "Mouse is over me!";
    });


    // Let box listen for mouseout
    box.addEventListener("mouseout", function () {
        box.innerHTML = "Mouse is out!";
    });
</script>
```

https://www.w3schools.com/js/js_events_mouse.asp



FORM VALIDATION

CLIENT-SIDE FORM VALIDATION

What is form validation?

- "This field is required" (You can't leave this field blank).
- "Please enter your phone number in the format xxx-xxxx" (A specific data format is required for it to be considered valid).
- "Please enter a valid email address" (the data you entered is not in the right format).
- "Your password needs to be between 8 and 30 characters long and contain one uppercase letter, one symbol, and a number." (A very specific data format is required for your data).

Client-side form validation - Learn web development | MDN (mozilla.org)

CLIENT-SIDE FORM VALIDATION

two different types of client-side validation

- Built-in form validation uses **HTML form validation** features, which we've discussed in many places throughout this module. This validation generally doesn't require much JavaScript. Built-in form validation has better performance than JavaScript, but it is not as customizable as JavaScript validation.

- **JavaScript validation** is coded using JavaScript. This validation is completely customizable, but you need to create it all (or use a library).

Client-side form validation - Learn web development | MDN (mozilla.org)

HTML FORM VALIDATION

The input **required** attribute specifies that an input field must be filled out before submitting the form.

The **required** attribute works with the following input types: **text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.**

```
<h2>JavaScript Validation</h2>
<form name="myForm" action="/action_page.php" onsubmit="return
validateForm()" method="post">

    Name: <input type="text" name="fname" required/>
    <input type="submit" value="Submit" />

</form>
```

https://www.w3schools.com/html/html_form_attributes.asp

JAVASCRIPT VALIDATION

```
<h2>JavaScript Validation</h2>
<form name="myForm" action="/action_page.php" onsubmit="return
validateForm()" method="post">
    Name: <input type="text" name="fname" />
    <input type="submit" value="Submit" />
</form>
<script>
    function validateForm() {
        let x = document.forms["myForm"]["fname"].value;
        if (x == "") {
            alert("Name must be filled out"); //alert message
            return false; //returns false, to prevent the form from being submitted
        }
    }
</script>
```

https://www.w3schools.com/js/js_validation.asp

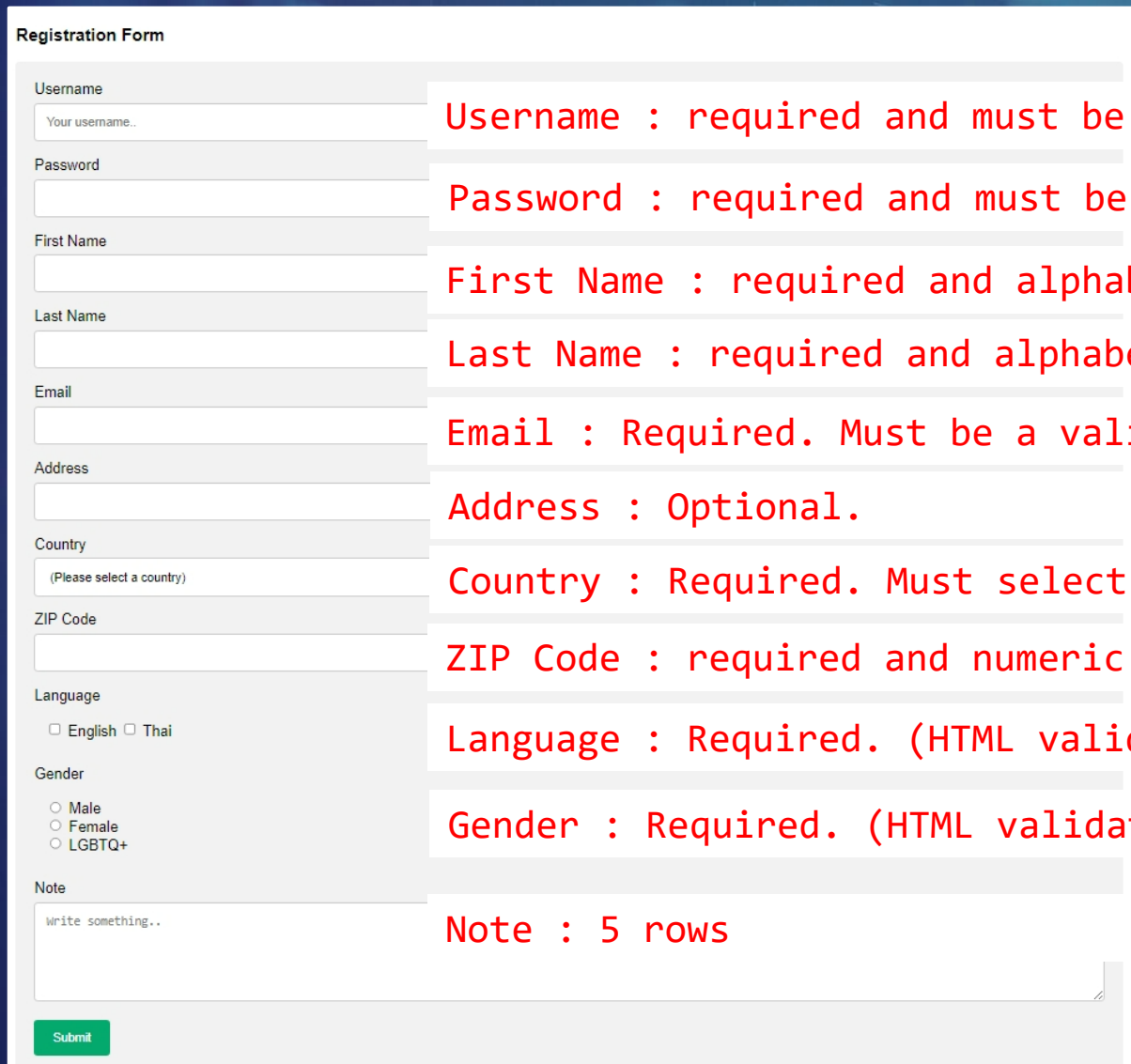
CONTACT FORM

How To Create a Contact Form with CSS (w3schools.com)



https://www.w3schools.com/howto/howto_css_contact_form.asp

REGISTRATION FORM



Username : required and must be of length 5-12.

Password : required and must be of length 6-16.

First Name : required and alphabet only.

Last Name : required and alphabet only.

Email : Required. Must be a valid email.

Address : Optional.

Country : Required. Must select a country.

ZIP Code : required and numeric only

Language : Required. (HTML validation)

Gender : Required. (HTML validation)

Note : 5 rows

REGISTRATION FORM

- ระบุ CSS ที่ไฟล์ style.css และ Javascript ที่ไฟล์ form_validation.js
- เปลี่ยนหัวข้อฟอร์ม เป็น “Registration Form”
- เพิ่ม submit event ไปที่ฟังก์ชัน
- เพิ่ม และ แก้ไขฟอร์มให้ถูกต้อง สมบูรณ์ และสวยงาม (ระบุทั้ง id และ name ของ input)
- เขียนฟังก์ชัน JavaScript เพื่อทำ Validation ตามกำหนด

https://www.w3schools.com/js/js_project_form_validation.asp

REGISTRATION FORM

Method length

```
function username_validation(uname, min, max) {
    let uname_len = uname.value.length;
    if (uname_len == 0 || uname_len >= max || uname_len < min) {
        alert("Username should not be empty / length be between " + min + "
to " + max);
        uname.focus();
        return false;
    }
    return true;
}
```

REGISTRATION FORM

Method match() - regular expression

```
function allLetter(inputtxt) {
    let letters = /^[A-Za-z]+$/;
    if (inputtxt.value.match(letters)) {
        return true;
    } else {
        alert(inputtxt.labels[0].innerHTML + ' must have alphabet characters
only');
        inputtxt.focus();
        return false;
    }
}
```

```
let numbers = /^[0-9]+$/;
let letters = /^[A-Za-z]+$/;
let letterNumber = /^[0-9a-zA-Z]+$/;
let mailformat = /^\w+([\.-]?\w+)*@\w+([\.-]?\w+)*(\.\w{2,3})+$/;
```

REGULAR EXPRESSION

Method match() - regular expression

https://regex101.com

https://regexr.com/

https://uibakery.io/regex-library/email

REGISTRATION FORM

list - selected

```
function countryselect(ucountry) {
    if (ucountry.value == "") {
        alert('Select your country from the list');
        ucountry.focus(); return false; }
    else {
        return true;
    }
}
```

ACTIVITY: STUDENT REGISTRATION

